

universität freiburg



7TH ENZYMEML WORKSHOP · RÜDESHEIM · HANDS-ON SESSION 2 · 16 SEPTEMBER 2026

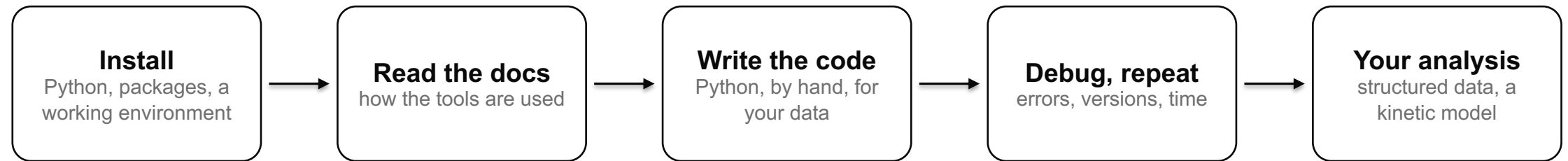
From measurement to kinetic modelling

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Last year the code was the hurdle

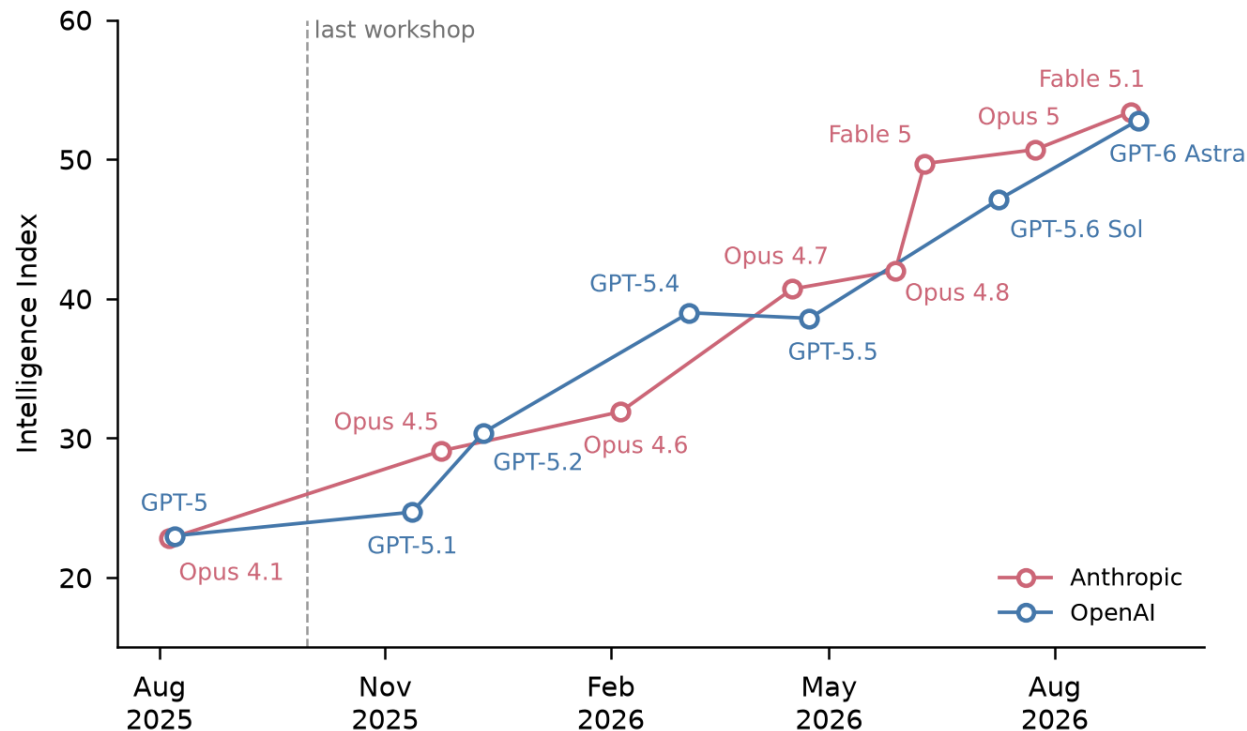
Previous hands-on sessions: we installed, read the docs and wrote the analysis code ourselves



- Failing installs and unfamiliar code: a hurdle for everyone who does not code daily
- So the data stayed unstructured, and the contemporary methods out of reach
- **Since then:** the models got better, and so did the harnesses that let them run tools

From 23 to 53 in one year

The two labs traded the lead several times; the biggest single step was Fable 5 in June 2026



- The Intelligence Index is an independent score from 0 to 100 that combines many tests of knowledge, reasoning, coding and agentic work, with every model tested the same way.
- Shown are the flagship models of Anthropic and OpenAI that anyone can use, each at its strongest public setting.

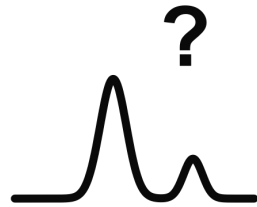
Artificial Analysis Intelligence Index v4.3, artificialanalysis.ai; release dates from the vendors

The assistant types, you decide



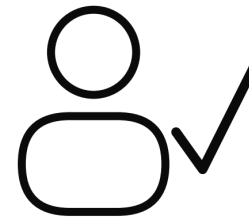
One page is enough

From a one-page card, today's models write working code for our tools



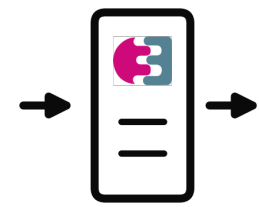
Blind to your data

Where the peak is, which model is right, when a result is wrong



You decide and check

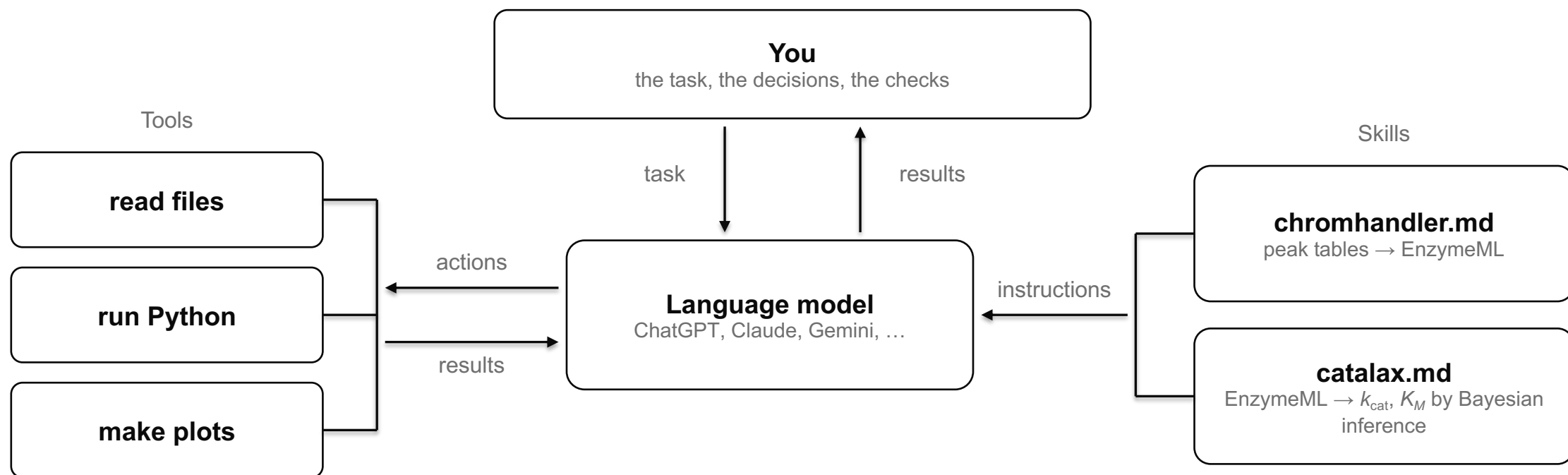
The decisions and the checks stay with the scientist; the assistant executes



EnzymeML in between

One document carries the data between the steps and the tools

Tools act, skills instruct, you decide

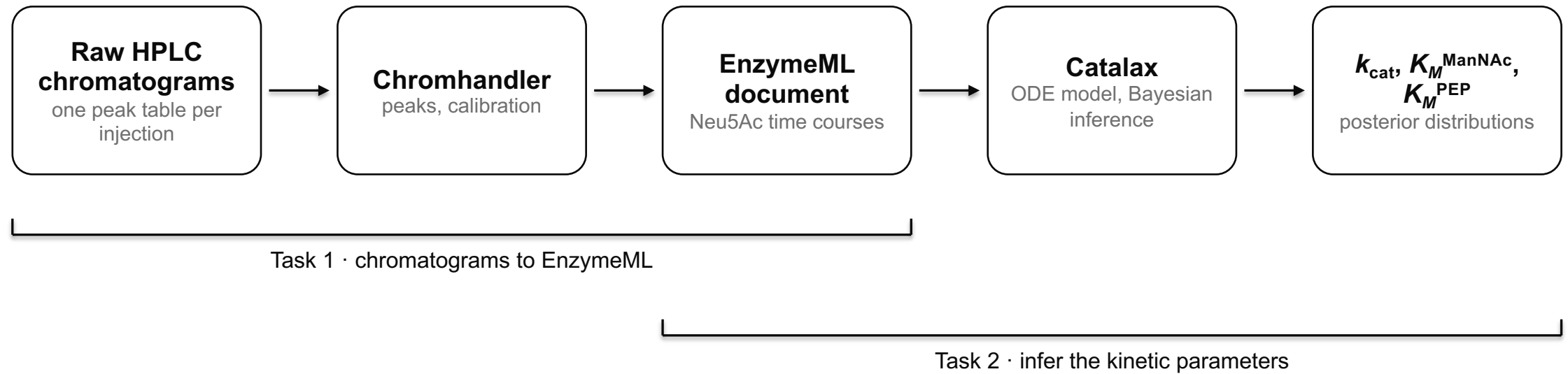


- Tools let the model read the data and run code; skills tell it how our libraries turn data into insight

WORKFLOW

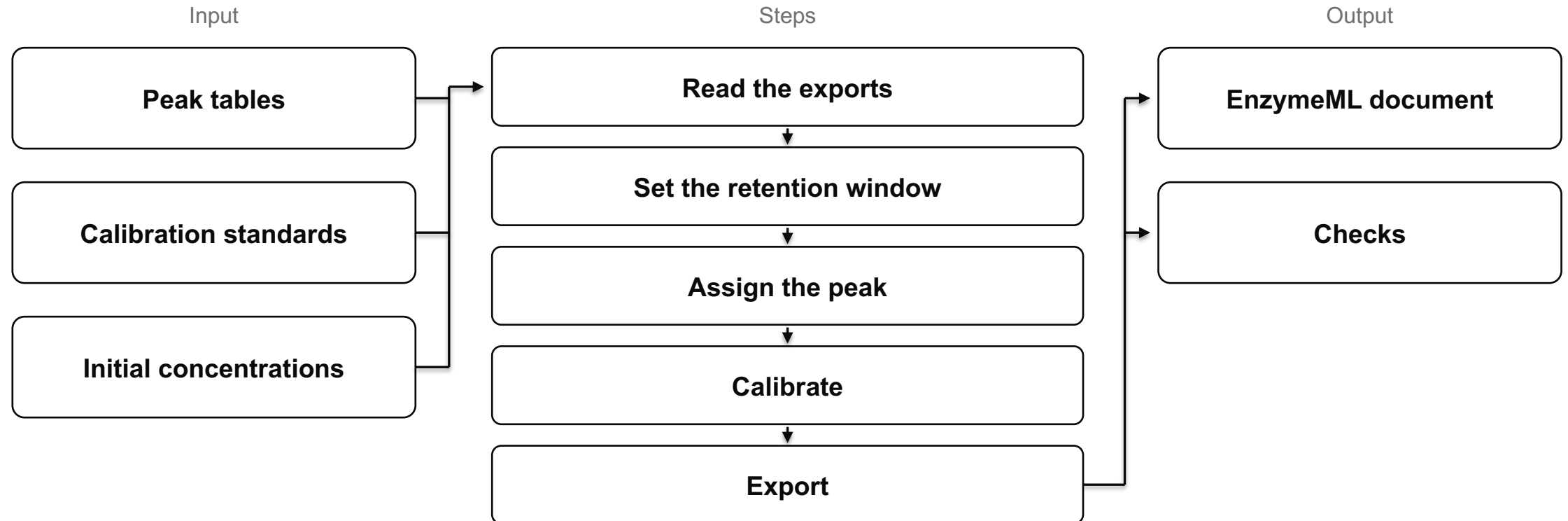
Two tasks, one document in between

Task 1 ends with the EnzymeML document; task 2 starts from it



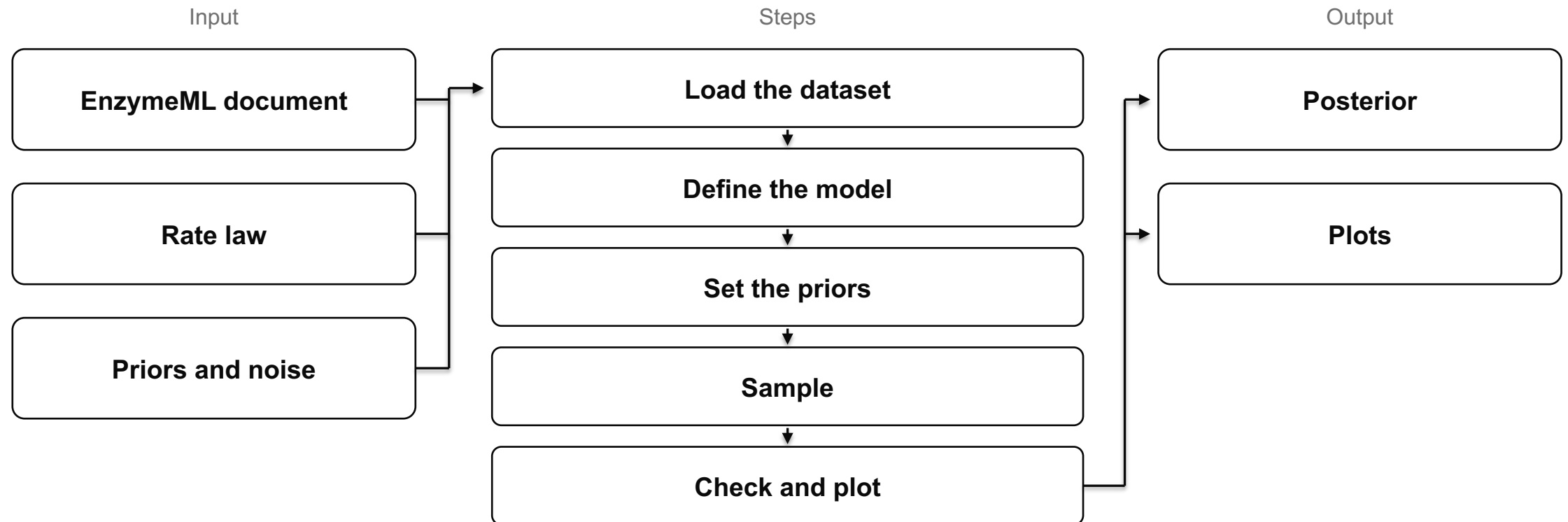
Chromhandler: from peak tables to concentrations

Reads the instrument's exports, assigns and calibrates one peak per compound, writes EnzymeML



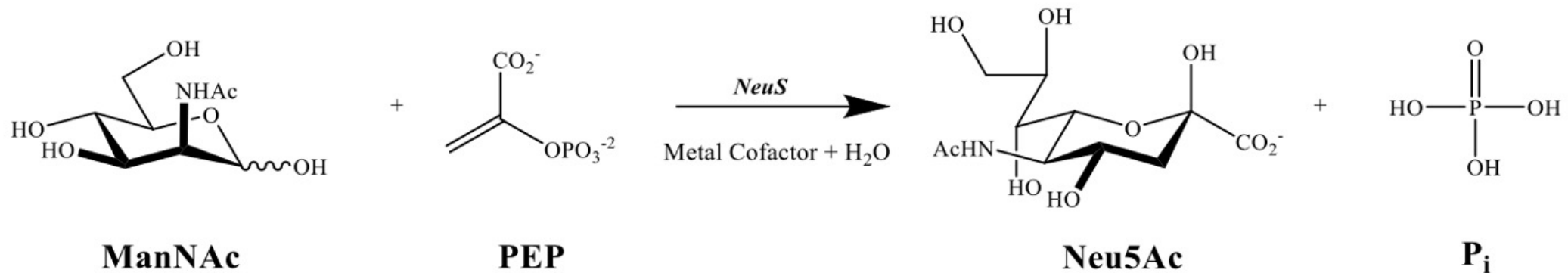
Catalax: from time courses to kinetic parameters

Builds the ODE model from a rate law and samples the posterior of its parameters



THE EXPERIMENT

ManNAc + PEP $\rightarrow$ Neu5Ac + Pi, followed by HPLC



- Reactions sampled over time and followed by HPLC-PDA at 215 nm
- What you get is the instrument's peak table per injection, not the raw trace
- Only the product Neu5Ac is quantified; calibration standards with known concentrations
- How many reactions, which conditions, how they were varied: that is in the data

Data: Çakar et al. (2024) Kinetic characterization of two neuraminic acid synthases and evaluation of their application potential. Appl. Microbiol. Biotechnol. 108, 446 · doi:10.1007/s00253-024-13277-1

Two tasks, one repository

Everything is in the README: tinyurl.com/enzymeml2026

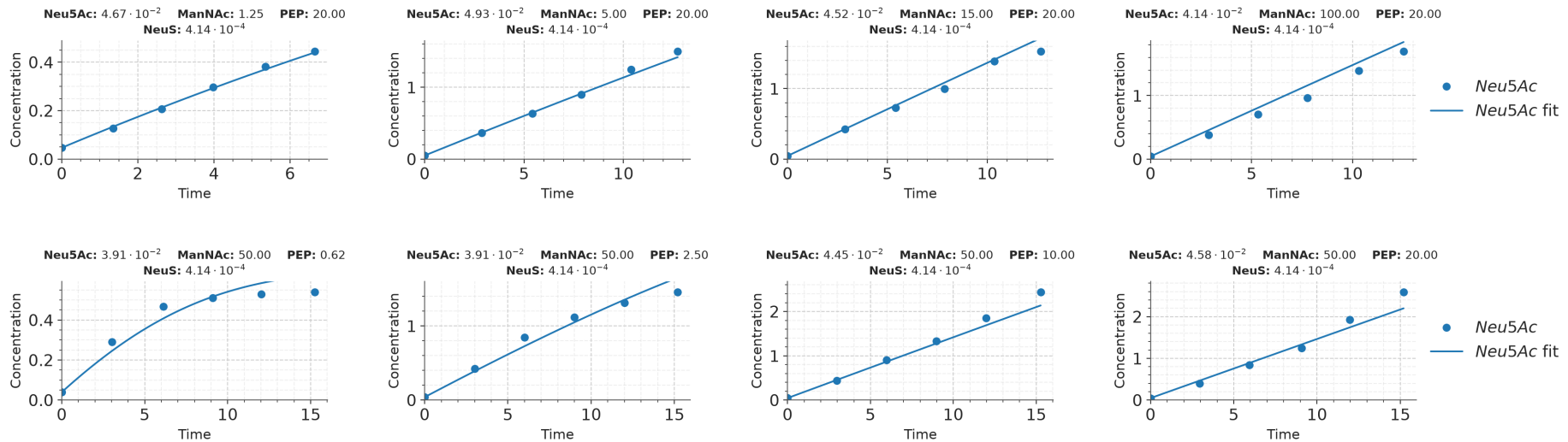
Tasks

- **Task 1 · chromatograms to EnzymeML:** plot the data, check the assay design, calibrate, export the document
- **Task 2 · kinetic parameters:** infer k_{cat} and K_M with their uncertainty and correlation; does the model describe the data?
- **At the end** we compare everybody's parameters

How to get started

- **Coding assistant on your computer:** give it the cards in briefs/
- **Chat assistant only:** the Colab starter notebook
- **No assistant that runs code?** team up
- **Details:** README, sections How to do it, Task 1, Task 2

The reference fit, and an open question



Parameter	Posterior mean $\pm$ sd
k_{cat}	$364 \pm 6 \text{ min}^{-1}$ (6.1 s^{-1} , 9.4 U/mg)
K_M^{ManNAc}	$1.53 \pm 0.19 \text{ mM}$
K_M^{PEP}	$0.59 \pm 0.07 \text{ mM}$

- One shared k_{cat} over-predicts the ManNAc series and under-predicts the PEP series
- The two series were measured on different days and differ by roughly 25 % in activity
- What would you do next?

What to take home

- A one-page card is enough for an assistant to drive our tools
- The decisions and the checks stay with the scientist
- The cards are reusable as SKILL.md or AGENTS.md in coding agents
- What did not work for you today is the most useful feedback for us

Thank you

tinyurl.com/enzymeml2026

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